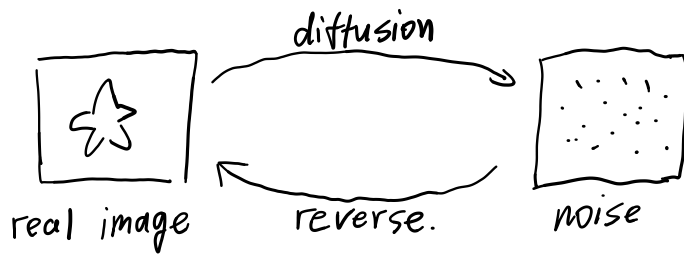
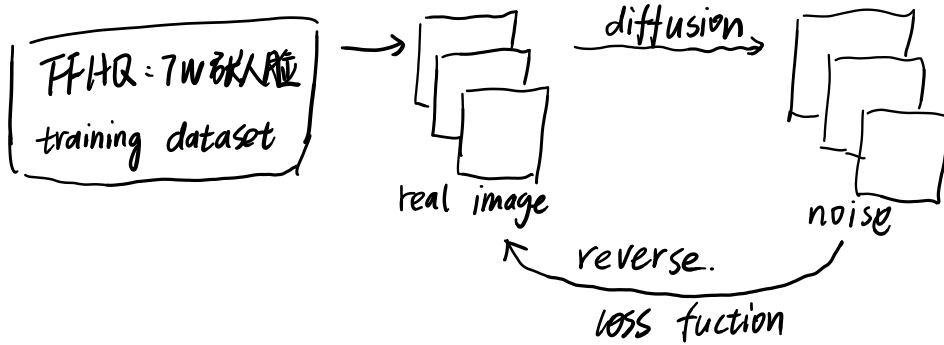


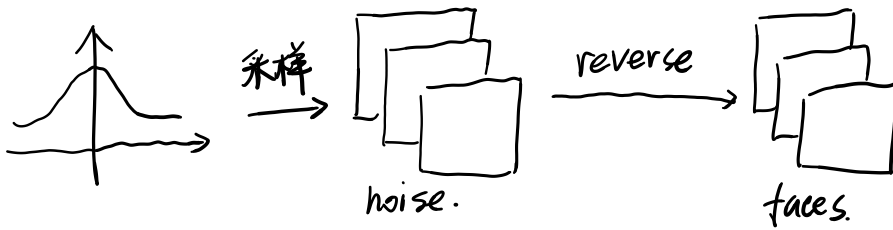
DDPM: Denoising Diffusion Probabilistic Models



① training phase



② inference phase



Diffusion 扩散阶段



$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} z_t \quad z_t \sim N(0, I)$$

$\beta: 10^{-4} \rightarrow 2 \times 10^{-2}$ 线性变化. $T=2000$

↓ 减少计算

$$x_t = f(x_0)$$

$$1 - \beta_t = \alpha_t \quad x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} z_t$$

$$z_t \sim N(0, I)$$

$$z_{t-1} \sim N(0, I)$$

← 代回

$$x_{t-1} = \sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} z_{t-1}$$

$$x_t = \sqrt{\alpha_t} (\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} z_{t-1}) + \sqrt{1 - \alpha_t} z_t$$

$$= \sqrt{\alpha_t \alpha_{t+1}} \cdot X_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t+1})} \cdot Z_{t-1} + \sqrt{1 - \alpha_t} \cdot Z_t$$

$$\sim N(0, \alpha_t - \alpha_t \alpha_{t+1}) \quad \sim N(0, 1 - \alpha_t)$$

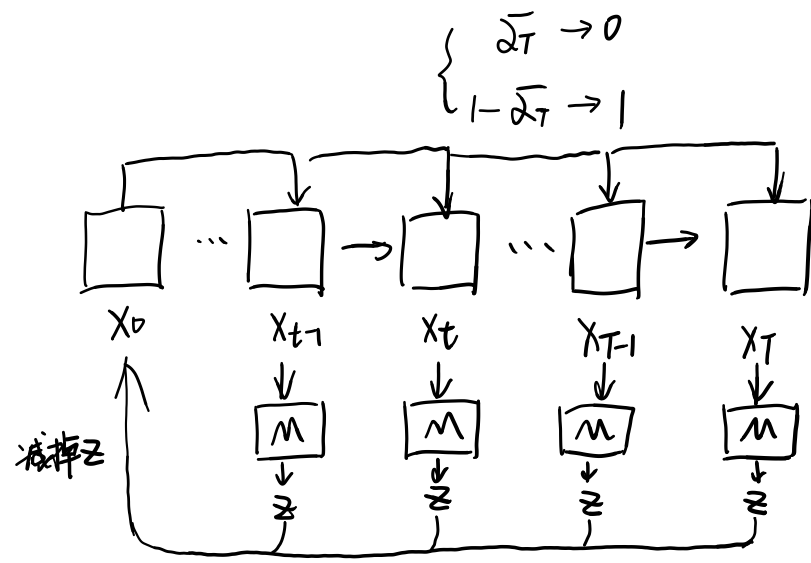
$$\sim N(0, 1 - \alpha_t \alpha_{t+1})$$

$$= \sqrt{\alpha_t \alpha_{t+1}} \cdot X_{t-2} + \sqrt{1 - \alpha_t \alpha_{t+1}} \cdot Z$$

$$X_t = \sqrt{\alpha_t \dots \alpha_1} \cdot X_0 + \sqrt{1 - \alpha_t \dots \alpha_1} \cdot Z \quad (Z \sim N(0, 1))$$

$$\star X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1 - \bar{\alpha}_t} Z \quad \bar{\alpha}_t = \prod_{i=0}^t \alpha_i$$

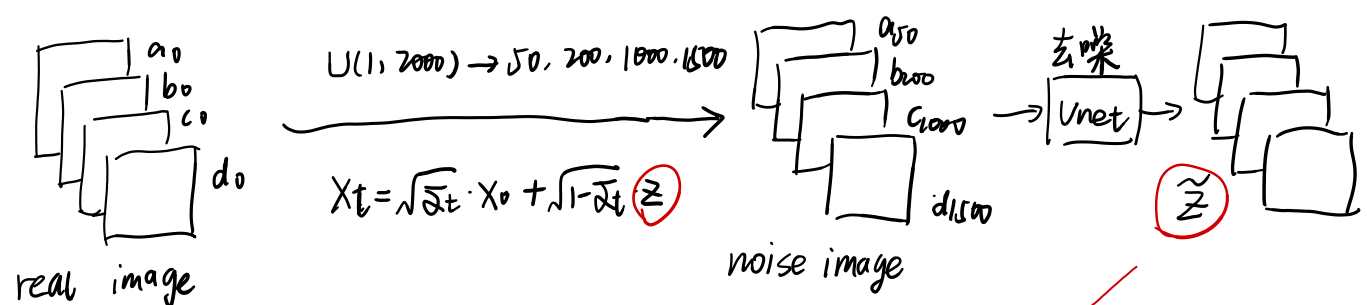
$$X_T = \sqrt{\bar{\alpha}_T} X_0 + \sqrt{1 - \bar{\alpha}_T} Z \rightarrow X_T \approx Z$$



Reverse 重建阶段

batch size = 4

$T = 2000$



$$X_t = \sqrt{\bar{\alpha}_t} \cdot X_0 + \sqrt{1 - \bar{\alpha}_t} \cdot Z$$

$$h(z, \hat{z}) \Rightarrow h_1 \text{ loss} / h_2 \text{ loss}$$

$$X_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} (X_t - \sqrt{1 - \bar{\alpha}_t} \cdot \hat{z})$$

一步完成 reverse 效果不好.

一层输出作为上一层输入，避免一步到位。

$$x_{t-1} = f(x_t, z)$$

? 误差不会叠加吗?

$$\hat{z} = \text{Unet}(x_t, t)$$

$$q(x_{t-1}|x_t) = \frac{q(x_t, x_{t-1})}{q(x_t)} = \frac{q(x_t|x_{t-1}) \cdot q(x_{t-1})}{q(x_t)}$$

$$\begin{aligned} x_t &= \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t} z_t, z_t \sim N(0, I) \Rightarrow \begin{cases} E(x_t|x_{t-1}) = E(\sqrt{\alpha_t} x_{t-1}) + 0 = \sqrt{\alpha_t} x_{t-1} \\ V(x_t|x_{t-1}) = 0 + V(\sqrt{1-\alpha_t} z_t) = (1-\alpha_t) I \end{cases} \Rightarrow \sim N(\sqrt{\alpha_t} x_{t-1}, (1-\alpha_t) I) \\ x_t &= \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} z, z \sim N(0, I) \Rightarrow \text{同理} \Rightarrow E(x_t|x_0) = \sqrt{\alpha_t} x_0 \quad V(x_t|x_0) = (1-\alpha_t) I \Rightarrow \sim N(\sqrt{\alpha_t} x_0, (1-\alpha_t) I) \end{aligned}$$

$$N(\mu, \sigma^2) \propto \exp\left(-\frac{1}{2} \cdot \frac{(x-\mu)^2}{\sigma^2}\right)$$

$$q(x_{t-1}|x_t) = \frac{q(x_t|x_{t-1}) q(x_{t-1})}{q(x_t)} \propto \exp\left\{-\frac{1}{2} \left(\frac{(x_t - \sqrt{\alpha_t} x_{t-1})^2}{1-\alpha_t} + \frac{(x_{t-1} - \sqrt{\alpha_{t-1}} x_0)^2}{1-\alpha_{t-1}} - \frac{(x_t - \sqrt{\alpha_t} x_0)^2}{1-\alpha_t} \right)\right\}$$

$$\propto \exp\left\{-\frac{1}{2} \left(\underbrace{\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}\right)}_A x_{t-1}^2 - 2 \underbrace{\left(\frac{\sqrt{\alpha_t} x_t}{\beta_t} + \frac{\sqrt{\alpha_{t-1}} x_0}{1-\alpha_{t-1}}\right)}_B x_{t-1} + \underbrace{?}_C \right)\right\}$$

$$\propto \exp\left\{-\frac{1}{2} (A x_{t-1}^2 + B x_{t-1} + C)\right\} = \exp\left\{-\frac{1}{2} \cdot A \left(x_{t-1} + \frac{B}{2A}\right)^2 + ?\right\}$$

$$\Rightarrow \begin{cases} \mu = -\frac{B}{2A} \\ \sigma^2 = \frac{1}{A} \end{cases} \quad \text{代入 A, B 得}$$

$$\sigma^2 = \frac{1}{A} = \frac{1}{\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\alpha_{t-1}}\right)} = \frac{1}{\frac{\alpha_t - \alpha_t \cdot \alpha_{t-1} + \beta_t}{\beta_t (1-\alpha_{t-1})}}$$

$$\because \alpha_t + \beta_t = 1$$

$$= \frac{1}{1 - \alpha_t \cdot \alpha_{t-1}} = \frac{1 - \alpha_{t-1} \cdot \beta_t}{1 - \alpha_t} \quad \text{化简}$$

$$\mu = -\frac{B}{2A} = \left(\frac{\sqrt{\alpha_t} x_t}{\beta_t} + \frac{\sqrt{\alpha_{t-1}} x_0}{1-\alpha_{t-1}}\right) \cdot \frac{1-\alpha_{t-1}}{1-\alpha_t} \cdot \beta_t$$

$$= \sqrt{\alpha_t} \cdot \frac{1 - \sqrt{\alpha_t}}{1 - \alpha_t} \cdot \underbrace{X_t}_{\text{已知}} + \frac{\sqrt{\alpha_t}}{1 - \alpha_t} \cdot \beta_t \cdot \underbrace{X_0}_{\text{未知, 只能通过估计的 } \tilde{z} \text{ 来计算}}$$

$$X_0 = \frac{1}{\sqrt{\alpha_t}} (X_t - \sqrt{1 - \alpha_t} \cdot \tilde{z})$$

代入可得:
$$m = \sqrt{\alpha_t} \cdot \frac{1 - \bar{\alpha}_t}{1 - \bar{\alpha}_t} \cdot x_t + \frac{\sqrt{\bar{\alpha}_t}}{1 - \bar{\alpha}_t} \cdot \beta_t \cdot \frac{1}{\sqrt{\alpha_t}} (x_t - \sqrt{1 - \bar{\alpha}_t} \cdot \tilde{z})$$
$$= \frac{x_t}{\sqrt{\alpha_t}} \left(\frac{\alpha_t - \bar{\alpha}_t + \beta_t}{1 - \bar{\alpha}_t} \right) - \frac{\tilde{z}}{\sqrt{\alpha_t}} \left(\frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \right)$$
$$= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \cdot \tilde{z} \right)$$

$$= \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \cdot \tilde{z} \right)$$

总结一下. $q(x_{t+1}|x_t) \sim N(\mu, \sigma^2) \Rightarrow x_t = \mu + \sigma^2 \cdot z$

$$\begin{cases} u = \frac{1}{\sqrt{2t}} \left(x_t - \frac{\beta_t}{\sqrt{1-2t}} \cdot \tilde{z} \right) \\ \sigma^2 = \frac{1-2t}{1-2t} \cdot \beta_t \end{cases}$$

$$\sigma^2 = \frac{1 - \alpha^2}{1 - \alpha} \cdot \beta_t$$

$$X \sim N(\mu, \sigma^2 I)$$

$$z \sim \mathcal{N}(0, I)$$

$$X = \mu + \sigma Z \sim N(\mu, \sigma^2)$$

用到重参数化技巧

$$X_t = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot z \right) + \sqrt{\frac{1-\alpha_t}{1-\beta_t}} \cdot \beta_t \cdot z \rightarrow \text{每次都加一个新噪声!}$$

$$\tilde{z} = V_{\text{net}}(X_t, t) \quad z \sim \mathcal{N}(0, I)$$

增加不确定性. 每次都在模拟布朗运动. 有确定的部分. 也有不确定的部分

